Model-Theoretic Quantifier Elimination

A Lean formalization over `mathlib4`

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Model Theory

What is Model Theory?

- Model theory is a branch of mathematical logic where we study mathematical structures by considering the first-order sentences true in those structures and the sets definable by first-order formulas.
- Given concrete mathematical structures, we can obtain new information about the structure using model-theoretic techniques.

Language

- In mathematical logic, we use first-order languages to describe mathematical structures.
- A $language$ $\mathcal{L}$ consists of:
  - a set of function symbols $\mathcal{F}$
  - a set of relation symbols $\mathcal{R}$
  - a set of constant symbols $\mathcal{C}$
- Examples:
  - The language of rings $\mathcal{L}_r = \{+, -, \cdot, 0, 1\}$
  - The language of orders $\mathcal{L}_o = \{<\}$

Structure

- A structure interprets the symbols of a language on a domain.
- Formally, an *$\mathcal{L}$-structure* $\mathcal{M}$ consists of:
  - a non-empty set M (the domain/universe),
  - for each function symbol $f \in \mathcal{F}$, a function $f^{\mathcal{M}} : \mathcal{M}^n \to \mathcal{M}$,
  - for each relation symbol $R \in \mathcal{R}$, a relation $R^{\mathcal{M}} \subseteq \mathcal{M}^n$,
  - for each constant symbol $c \in \mathcal{C}$, an element $c^{\mathcal{M}} \in \mathcal{M}$.
- Examples: $(\mathbb{N}, 0, 1, +, \cdot)$, $(\mathbb{Q}, \leq)$.

Terms

- An $\mathcal{L}$-*term* is:
  - a constant symbol c,
  - a variable v_i
  - or a function symbol f applied to terms $t_1, \ldots, t_n$.

Formula and Sentence

- An *atomic $\mathcal{L}$-formula* is a relation symbol applied to terms, or an equality between terms.
- *$\mathcal{L}$-formulas* are built from atomic formulas using logical connectives and quantifiers.
- An *$\mathcal{L}$-sentence* is a formula with no free variables.
- Example: $\exists z\,(x < z \wedge z < y)$ is formula but not a sentence, as it has free variables x and y.

Satisfaction

- We say that an $\mathcal{M}$ *satisfies* a formula φ, written $\mathcal{M} \models \varphi$, if φ holds in $\mathcal{M}$.
- Formally, the satisfaction relation is defined inductively on the structure of formulas.
- Example: $(\mathbb{Q}, <) \models \forall x \forall y \, (x < y \implies \exists z \, x < z \wedge z < y)$.

Theory and Model

- An $\mathcal{L}$-*theory* is a set of $\mathcal{L}$-sentences.
- We say that a structure $\mathcal{M}$ is a *model* of a theory T, written $\mathcal{M} \models T$, if $\mathcal{M}$ satisfies every sentence in T.
- A theory is *satisfiable* if it has a model.
- Moreover, we say that a theory T satisfies a sentence φ, written $T \models \varphi$, if every model of T satisfies φ.

Quantifier Elimination

- A theory T has *quantifier elimination* if for every formula φ, there is a quantifier-free formula ψ such that

$$T \models \varphi \leftrightarrow \psi$$

Dense Linear Orders without Endpoints

- Let $\mathcal{L}_o = \{<\}$ be the language of orders.
- Let DLO be the $\mathcal{L}_o$-theory consisting of the following sentences:

$\forall x \neg(x < x)$	(Irreflexivity)
$\forall x \forall y \forall z\,(x < y \wedge y < z \to x < z)$	(Transitivity)
$\forall x \forall y\,(x < y \vee x = y \vee y < x)$	(Linearity)
$\forall x \forall y\,(x < y \to \exists z\,(x < z \wedge z < y))$	(Density)
$\forall x \exists y \exists z (y < x \wedge x < z)$	(No endpoints)

- Examples of models of DLO: $(\mathbb{Q}, <)$, $(\mathbb{R}, <)$, $((0,1), <)$ with the usual order.

Does DLO have Quantifier Elimination?

- DLO has quantifier elimination.
- Every formula in the language $\{<\}$ is equivalent over DLO to a quantifier-free formula.
- Examples:

$$\exists z\,(x < z \wedge z < y) \iff x < y$$

$$\exists z\,(x < z \wedge w < z \wedge z < y) \iff x < y \wedge w < y$$

$$\exists z\,\big(x < z \wedge z < y \wedge \neg(z < w)\big) \iff x < y \wedge (w < y \vee w = y)$$

Type Theory

Judgements

- $a : A$ is a judgement: a has type A.
- Every expression has a type.
- Definitional equality is written $a \equiv b$.

Propositions as Types

- A proposition is viewed as a type of evidence.
- A proof of P is a term $p : P$.
- To prove a theorem is to construct an object of the corresponding type.

Constructive Reasoning

- A proof of $\exists x, P(x)$ gives a witness and evidence.
- A proof of $P \vee Q$ chooses one side and proves it.
- The law of excluded middle, $P \vee \neg P$, is not built in constructively.
- Lean allows classical reasoning when explicitly invoked.

Curry–Howard Isomorphism

$$
\begin{aligned}
[\![P \Rightarrow Q]\!] &\equiv [\![P]\!] \to [\![Q]\!] \\
[\![P \wedge Q]\!] &\equiv [\![P]\!] \times [\![Q]\!] \\
[\![\mathrm{True}]\!] &\equiv 1 \\
[\![P \vee Q]\!] &\equiv [\![P]\!] + [\![Q]\!] \\
[\![\mathrm{False}]\!] &\equiv 0 \\
[\![\forall x : A.\, P]\!] &\equiv \Pi x : A.\, [\![P]\!] \\
[\![\exists x : A.\, P]\!] &\equiv \Sigma x : A.\, [\![P]\!]
\end{aligned}
$$

Formalization in Lean

```
/-- A formula is equivalent to a quantifier-free formula over a theory. -/
def IsQFEquivalent (T : L.Theory) {α : Type*} (φ : L.Formula α) : Prop :=
  ∃ ψ : L.Formula α, ψ.IsQF ∧ φ ⇔[T] ψ

/-- A theory has quantifier elimination if every formula is equivalent,
over the theory, to a quantifier-free formula. -/
def HasQuantifierElimination (T : L.Theory) : Prop :=
  ∀ {α : Type} (φ : L.Formula α), T.IsQFEquivalent φ
```

```
/-- The theory of dense linear orders without endpoints
has quantifier elimination. -/
theorem dlo_hasQuantifierElimination :
    Language.order.dlo.HasQuantifierElimination := by
  ...
```

Thank you for listening!

Scan the QR code to checkout our PR in `mathlib4` Github repository!